

The mirror test: a $Z/2$ group action on the skeleton

Bilateral reflection is a two-element group acting on the skeleton. Read left to right.

The anatomical mirror M acts on the skeleton

M negates x and swaps left/right, sending x to Mx .



A $Z/2$ group: $G = \{I, M\}$

I = identity (do nothing)

M = the anatomical mirror

$M(Mx) = x$ (involution: twice = identity)

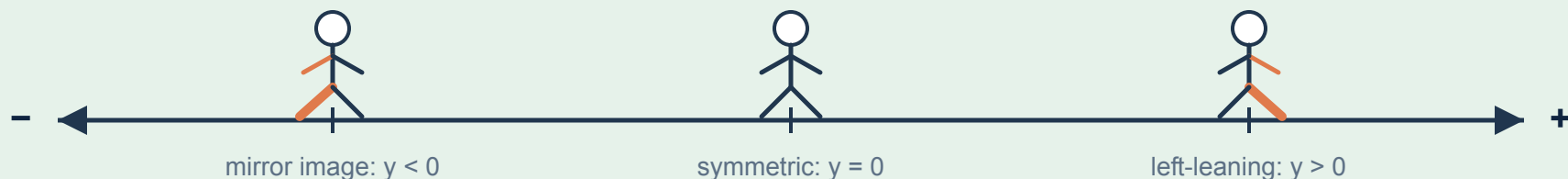
M swaps 6 left/right landmark pairs:

shoulder · hip · knee

ankle · heel · foot-index

The signed laterality readout y is antisymmetric under the mirror

$y = \sum$ over the 6 left/right pairs (left temporal-excursion - right). Under M the sign flips: $y(Mx) = -y(x)$.



Ideal reflection-equivariance: mirrored output = - original output (slope -1).